**Arab Academy for Science, Technology and Maritime Transport**

College of Computing and Information Technology

Software Engineering Department

CSE4203 Principles of Software Architecture

# Tamkeen

Inclusive Employment and 5% Quota Compliance Platform

## Milestone 1: Proposal and Architectural Requirements

Team Name: Success Enablers

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# 1. System overview

**The problem.** Under Article 22 of Law No. 10 of 2018, any employer with 20 or more workers has to give at least 5% of its jobs to persons with disabilities (PwDs) [1], [3]. In practice the law is poorly enforced. Human Rights Watch reported this in June 2026, and by the end of 2025 only about 10% of PwDs (1.3 million people) had the Integrated Services Card that proves their status [3]. Some companies get around the quota with "paper jobs": they pay a PwD to stay at home so they can count them on the books [4]. Job seekers have their own problems. Most job sites don't work well with a screen reader, and many forms ask for a diagnosis.

**Who uses it.** Job seekers with visual, hearing, mobility or cognitive disabilities, sometimes with a job coach helping them. HR staff at companies. And verifiers on our side, who check that companies are real and that a person's disability card is valid.

**What it does.** Tamkeen is a job platform for PwDs, in Arabic and English. A job seeker's profile lists their skills and what they need to work well, such as a screen reader, step-free access or written instructions. We never ask for a diagnosis. Companies post jobs and say which accommodations they offer, and Tamkeen matches people to jobs that fit. When a company hires a verified PwD through the platform, its quota dashboard updates. A company with 120 employees, for example, needs 6 PwDs. If it has 2 verified, the dashboard shows a gap of 4.

**Why it is architecturally interesting.** On features alone, Tamkeen is a job board where people post, search and apply, and a single monolithic block could do that. What makes the architecture hard is a set of quality attributes that conflict with each other:

| Trade-off | Where it shows up in Tamkeen |
|---|---|
| **Security ⇄ Usability** | The usual protections, like image CAPTCHAs, short session time-outs and extra login steps, lock out screen-reader users and users with cognitive disabilities. So we have to get security from things the user never sees, such as server-side authorization, rate limiting and audit trails. |
| **Security (confidentiality) ⇄ Usability for employers** | Law 151/2020 treats disability data as sensitive, so we need explicit consent that the user can withdraw [2]. Employers want details so they can plan accommodations, while candidates want to decide what is shared. That means the proof of disability status has to live apart from the profile employers see. |
| **Availability ⇄ Cost** | We run on free cloud tiers, and the free API server sleeps when nobody uses it. A job seeker still shouldn't lose an application because of that, and we can't pay for always-on backup servers. |

Two other things shape the design. Many companies share one database, and none of them may see another's data. The quota rules come from the law and could change, so we want to update them without editing code.

## 2. Stakeholders and concerns

Each stakeholder cares about a different quality, which is why the scenarios in section 5 come from several of them. The last column names the quality attribute that matters most to each group.

| Stakeholder | Who | Key concerns | Main quality attribute |
|---|---|---|---|
| PwD job seekers | Visual, hearing, mobility and cognitive disabilities | Works with their assistive technology; never asked for a diagnosis; control over who sees disability data; real jobs, not paper jobs | Usability, Security |
| Job coaches / family helpers | Parents, NGO job coaches | Can help without taking over; clear, revocable permission | Usability, Security |
| Employer HR recruiters | Recruiters, HR managers | Fast job posting; understanding accommodations; fast candidate search | Usability, Performance |
| Employer management | Owners, compliance officers | Correct quota figures; proof of compliance with Article 22; avoiding fines of 10,000 to 30,000 EGP and possible imprisonment [3] | Security (integrity) |
| Admins / verifiers | Platform team or a partner NGO | Fast verification queue; stopping fake companies; not keeping documents longer than needed | Security, Usability |
| Developers | The 5-person team | Simple stack, low coupling and high cohesion, easy to test, delivered on time | Modifiability, Testability |
| Operators | The team, as part-time operators | Free-tier limits, monitoring, easy redeploy and rollback | Availability |
| Disability regulator | National Council for Persons with Disabilities; Ministry of Social Solidarity | Quota computed per Law 10/2018; real employment; Integrated Services Card respected | Modifiability (rules), Security |
| Data-protection regulator | Personal Data Protection Center | Law 151/2020: consent, data minimization, security, transfers outside Egypt | Security |
| Wider Tamkeen platform | Graduation-project team and supervisor | Shared sign-in, clean API for the mobile app, consistent accessibility settings | Interoperability |

# 3. System context (C4 Level 1)

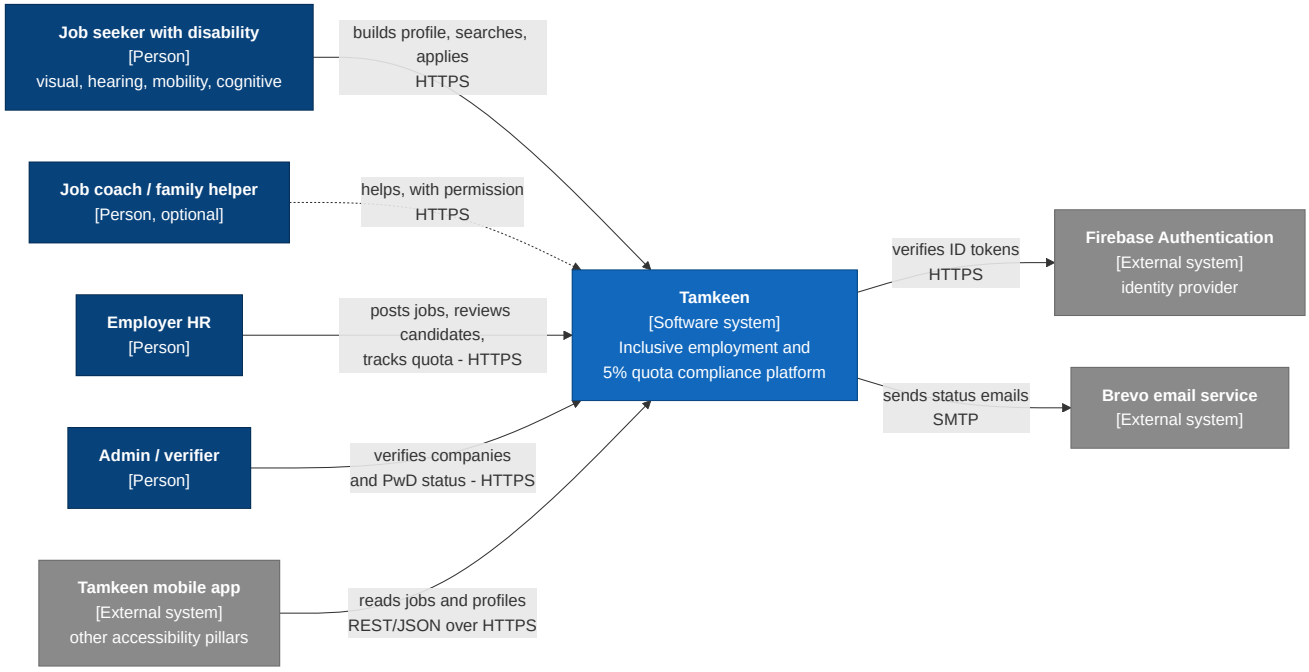



Figure 1. Tamkeen system context. Source file: diagrams/context.mmd

Person (user role) The system being built (one box) External system

Solid arrow: interaction, labelled with its purpose and protocol Dashed arrow: optional (Could priority)

| External system | Role in the system | If it is unavailable |
|---|---|---|
| Firebase Authentication | Identity provider: one account across the Tamkeen platform; Tamkeen verifies its ID tokens | No new sign-ins; active sessions continue until tokens expire |
| Brevo email service (SMTP) | Application-status emails; the main contact channel for deaf users | Emails queue and retry; status stays visible in the app |
| Tamkeen mobile app | Shows jobs inside the wider platform through Tamkeen's REST API | No effect on the web app |

# 4. Functional requirements

We wrote 15 user stories and ranked them with MoSCoW. In the M3 column, ✓ means we will build it in the Milestone 3 prototype, Stretch means we will build it if there is time, and × means it is not in the prototype. The Related QA column links each story to the quality attribute it affects most.

| ID | User story | MoSCoW | M3 | Related QA |
|---|---|---|---|---|
| FR-01 | As any user, I sign in with my Tamkeen account and choose my role (job seeker, employer, verifier). | Must | ✓ | Security, Interoperability |
| FR-02 | As a job seeker, I set language (Arabic/English), text size, contrast theme and easy-read mode before anything else. | Must | ✓ | Usability |
| FR-03 | As a job seeker, I add skills, experience and work needs (e.g., screen reader, step-free access) without stating a diagnosis. | Must | ✓ | Usability, Security |
| FR-04 | As a job seeker, I give explicit consent before disability-related data is stored, see who can view it, and withdraw or delete it at any time. | Must | ✓ | Security |
| FR-05 | As an employer, I register my company with its headcount and post jobs that list the accommodations offered. | Must | ✓ | Usability |
| FR-06 | As a job seeker, I search jobs by keyword, city and remote option, and filter by accommodations. | Must | ✓ | Performance |
| FR-07 | As a job seeker, I apply to a job, choose which work needs to share with that employer, and track the status. | Must | ✓ | Availability, Security |
| FR-08 | As an employer, I see a quota dashboard: required PwD headcount, current verified PwD staff, gap and percentage. | Must | ✓ | Security, Modifiability |
| FR-09 | As a verifier, I approve or reject companies and PwD status (Integrated Services Card upload); the file is deleted after my decision. | Must | ✓ (simplified) | Security |
| FR-10 | As a job seeker, I see a match score with plain reasons, e.g., "offers screen-reader software; matches 4 of 5 skills". | Should | Stretch | Usability |
| FR-11 | As a job seeker, I receive an email whenever my application status changes. | Should | Stretch | Interoperability |
| FR-12 | As an employer, I export a monthly quota compliance report as PDF. | Should | × | Modifiability |
| FR-13 | As a job seeker, I generate a screen-reader-friendly CV (HTML/PDF) from my profile. | Could | × | Usability |
| FR-14 | As an employee, I anonymously review whether my job is real; companies receive a "reality score". | Could | × | Security |
| FR-15 | As a job seeker, I connect with others, endorse skills and post to a social feed. | Won't | × | — |

> The system has three main user roles: job seeker, employer HR and admin/verifier, plus an optional job coach. The prototype will cover all nine Must stories, so a user can go from signing in all the way to an updated quota dashboard.

# 5. Quality attribute scenarios

We wrote eight scenarios in the six-part form, covering six quality attributes: usability, security, availability, performance, modifiability and interoperability. We count accessibility under usability and privacy under security (confidentiality). Each response measure is something we can observe and test on the Milestone 3 prototype.

### QAS-1 · Usability (accessibility): a blind user applies for a job

| | |
|---|---|
| **Source** | Blind job seeker (end user) using the NVDA screen reader and a keyboard only |
| **Stimulus** | Wants to use the system efficiently: complete onboarding, build a profile and apply to a job |
| **Artifact** | Web user interface: the 6 core pages (sign-in, preferences, profile, search, job details, apply) |
| **Environment** | First use, normal operation, Arabic interface |
| **Response** | Every control is reachable by keyboard, labelled and announced in the correct language; feedback is spoken after each step; no CAPTCHA and no form time-out |
| **Response measure** | 0 critical or serious axe-core violations on all 6 pages; Lighthouse accessibility score ≥ 95 per page; task completed without sighted help in ≤ 15 minutes in a walkthrough |

**Candidate tactics (Lecture 2):** Support system initiative (provide feedback and assistance); support user initiative (pause / resume, cancel).

### QAS-2 · Usability: a user with an intellectual disability applies in easy-read mode

| | |
|---|---|
| **Source** | Job seeker with an intellectual disability (novice end user), helper nearby |
| **Stimulus** | Wants to find a job near home and apply |
| **Artifact** | Web user interface in easy-read mode |
| **Environment** | Normal operation |
| **Response** | One task per screen, short sentences with icons, progress saved after every step, no time limits |
| **Response measure** | Apply flow takes ≤ 6 screens; completed in ≤ 10 minutes in a walkthrough; ≥ 4 of 5 test users rate it "easy" |

**Candidate tactics (Lecture 2):** Aggregate related actions; pause / resume; maintain a user model (easy-read preference stored once, applied everywhere).

### QAS-3 · Security (confidentiality): an employer views an applicant

| | |
|---|---|
| **Source** | Authenticated employer HR user (known user) |
| **Stimulus** | Attack to display information: tries to view an applicant's profile, including disability-related data |
| **Artifact** | Applications service and the sensitive-data store |
| **Environment** | Online, normal operation |
| **Response** | Returns only skills and the work needs the candidate chose to share for this application; disability status details, card files and consent records are never returned |
| **Response measure** | 100% of employer-facing endpoints pass an automated test confirming sensitive fields are absent; 100% of stored sensitive records link to a consent record; after consent is withdrawn, data is hidden immediately and deleted within 24 hours (Law 151/2020) |

**Candidate tactics (Lecture 2):** Resist attacks: authorize actors, limit exposure, encrypt data, separate entities; recover: maintain audit trail.

### QAS-4 · Security (authorization): a company tries to read another company's data

| | |
|---|---|
| **Source** | Authenticated HR user of Company A (malicious or faulty client) |
| **Stimulus** | Attack to access information: tries to read Company B's quota data or applicants by changing an ID in the request |
| **Artifact** | API authorization layer |
| **Environment** | Online, normal operation |
| **Response** | Request is denied (HTTP 403), no data is returned, and the attempt is logged with user ID and target |
| **Response measure** | 100% of at least 20 cross-company test cases denied; 0 records from another company in any response; each denial logged within 1 second |

**Candidate tactics (Lecture 2):** Resist attacks: authorize actors, limit access; detect attacks: detect intrusion; react: lock the account and inform actors after repeated attempts; recover: maintain audit trail.

### QAS-5 · Availability: the server is asleep when someone applies

| | |
|---|---|
| **Source** | Infrastructure (free-tier hosting platform) |
| **Stimulus** | Fault (omission): the API instance is asleep or restarting when a job seeker submits an application |
| **Artifact** | Web client and API process |
| **Environment** | Degraded mode: free-tier hosting, low traffic |
| **Response** | Fault is detected; the client keeps the draft on the device, shows an accessible "connecting, your answers are saved" message and retries; the server ignores duplicate submissions |
| **Response measure** | 0 lost and 0 duplicated applications in 20 forced-restart tests; submission completes ≤ 90 seconds after the API wakes; monthly uptime ≥ 99% on a free uptime monitor |

**Candidate tactics (Lecture 2):** Detect faults: ping/echo, heartbeat (uptime monitor); recover: retry, degradation; prevent: transactions (idempotent submit).

### QAS-6 · Performance: job search under load

| | |
|---|---|
| **Source** | Job seekers (external) |
| **Stimulus** | 100 concurrent users submit filtered job searches (stochastic arrival) |
| **Artifact** | Job search service and database |
| **Environment** | Normal operation, free-tier hosting, API awake |
| **Response** | Matching jobs are returned, paginated, with the user's accessibility filters applied |
| **Response measure** | 95% of requests answered in < 1.5 seconds; 0% failed requests during a 5-minute load test |

**Candidate tactics (Lecture 2):** Control resource demand: limit event response (pagination), increase efficiency (database indexes); manage resources: maintain multiple copies of data (cache).

### QAS-7 · Modifiability: the quota rule changes

| | |
|---|---|
| **Source** | System administrator |
| **Stimulus** | Wishes to modify the quota rule (percentage, threshold or rounding) after a change in the law |
| **Artifact** | Quota engine rule configuration (data, not code) |
| **Environment** | Run time, production |
| **Response** | New rule version is added, applies from its effective date, past months keep their original results, with no side effects |
| **Response measure** | Change completed in ≤ 10 minutes; 0 code changes and 0 redeployments; 100% of quota unit tests pass for both rule versions |

**Candidate tactics (Lecture 2):** Defer binding time: configuration files / runtime registration; increase cohesion: increase semantic coherence (one quota module owns all rule logic).

### QAS-8 · Interoperability: the Tamkeen mobile app asks for job listings

| | |
|---|---|
| **Source** | Tamkeen mobile app (a system) |
| **Stimulus** | Request to exchange information: asks for job listings and the user's application status, using a Firebase ID token |
| **Artifact** | Tamkeen REST API and the mobile app (the systems that interoperate) |
| **Environment** | Systems known before run time, normal operation |
| **Response** | Valid requests get JSON that matches the published OpenAPI contract; requests with a missing or expired token are rejected with HTTP 401 and logged |
| **Response measure** | 100% of 50 contract-test requests correctly processed; 100% of 10 invalid-token requests correctly rejected and logged |

**Candidate tactics (Lecture 2):** Manage interfaces: tailor the interface (one documented REST/JSON contract); detect and interpret errors from the other party (clear error codes).

# 6. Constraints

| ID | Type | Constraint |
|---|---|---|
| C-T1 | Technical | MERN stack (MongoDB, Express, React, Node.js); JavaScript throughout. |
| C-T2 | Technical | Sign-in must use the Tamkeen platform's shared identity provider (Firebase Authentication). |
| C-T3 | Technical | Must run on free tiers only: MongoDB Atlas free cluster, a free web-service host for the API (sleeps when idle), free static hosting for the web app, a free email tier. |
| C-T4 | Technical | Must meet WCAG 2.1 Level AA [5] in Arabic (right-to-left) and English. |
| C-T5 | Technical | Must expose a REST/JSON API, documented with OpenAPI, for the Tamkeen mobile app. |
| C-T6 | Technical | No public government API exists to check the Integrated Services Card, so verification is manual. |
| C-B1 | Business / legal | Law 10/2018, Article 22: employers with ≥ 20 workers must employ ≥ 5% PwDs [1], [3]. The law does not specify rounding; we round up (30 workers → 2 required). |
| C-B2 | Business / legal | Law 151/2020: disability data is sensitive personal data; explicit consent, right to withdraw, limits on transfer outside Egypt [2]. Free cloud tiers are hosted abroad, so the prototype uses synthetic data only. |
| C-B3 | Business | No payments or fees in this phase. |
| C-B4 | Business | Never ask for a diagnosis; collect work needs instead. |
| C-O1 | Organizational | Five part-time students with beginner-to-intermediate MERN skills; no dedicated DevOps member. |
| C-O2 | Organizational | Deadlines: M1 on 23 October 2026; project discussion in university weeks 12 and 13 (5 to 18 December 2026). |
| C-O3 | Organizational | All work submitted through GitHub; documents as PDF. |
| C-O4 | Organizational | The system is also the employment pillar of the team's Tamkeen graduation project and must fit its shared sign-in and settings. |

# 7. Architecturally significant requirements

We used the rule from Lecture 1: a requirement is architecturally significant when a quality attribute drives it and the decision behind it would be expensive to change later. Each scenario in the tree is rated (Importance, Difficulty). Importance is how much it matters to the customer (business value). Difficulty is how hard it is for us to achieve (architectural impact).

## Utility tree

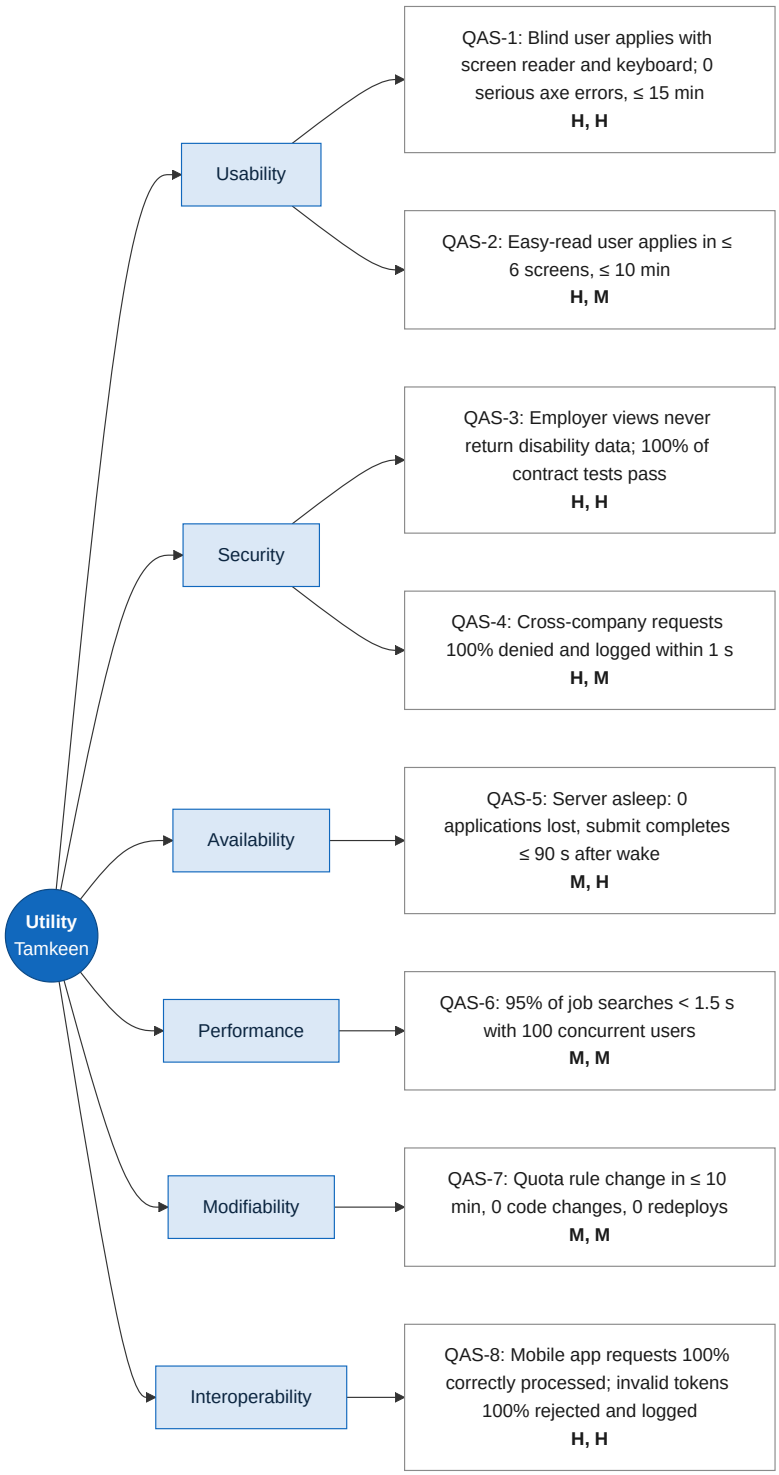



Figure 2. Utility tree for Tamkeen: quality attribute → scenario → (Importance, Difficulty). Source file: diagrams/utility-tree.mmd

## Prioritized requirements (sections 4 to 6)

| Rank | Requirement | Quality attribute | Business value | Arch. impact | Why it is architecturally significant |
|---|---|---|---|---|---|
| **1** | QAS-3, FR-04, C-B2: confidentiality of disability data | Security | **H** | **H** | Needs a separate, encrypted store for sensitive data, consent records, and different API responses per role. Adding this after the data model exists would mean changing every module that reads profiles. |
| **2** | QAS-1, QAS-2, FR-02, C-T4: accessible use for all disability types | Usability | **H** | **H** | Needs one shared library of accessible UI components, Arabic and English (with RTL) handled once at the app shell, and user preferences stored in one place. Fixing accessibility screen by screen later costs much more. |
| **3** | QAS-8, FR-01, C-T2, C-T5: shared Tamkeen identity, API-first | Interoperability, Security | **H** | **H** | Needs an external identity provider, a token check on every request, users identified by an external ID, and a documented API that the mobile app can also call. |
| **4** | QAS-4, FR-08: company isolation | Security | **H** | **M** | Needs one authorization layer that limits every query to the user's own company. Checks spread across handlers are easy to forget. |
| **5** | QAS-5, C-T3: availability on free hosting | Availability | **M** | **H** | Needs drafts saved on the client, automatic retry, and submit endpoints that are safe to call twice, since we can't keep the server awake. |
| 6 | QAS-6, FR-06: search response time | Performance | **M** | **M** | Needs indexes, pagination and caching; can be tuned later without restructuring. |
| 7 | QAS-7, C-B1: configurable quota rules | Modifiability | **M** | **M** | Rules are stored as versioned data in one quota module. |
| 8 | FR-09, C-T6: manual verification | Security | **H** | **M** | Needs a verification queue, short-lived file storage and an audit trail. |
| 9 | FR-10: explainable matching | Usability | **M** | **L** | Self-contained and fits in any reasonable structure. |
| 10 | FR-11, FR-12: email, PDF report | Interoperability | **M** | **L** | Can sit behind an interface and be added later. |

Ranks 1 to 5 are our ASRs, and they will drive the design in Milestone 2. We still have to meet ranks 6 to 10, but they will fit inside whatever structure the top five lead to.

## Contribution table

| Member | Contribution | Reviewed |
|---|---|---|
| Zaid Mohamed Abdelrahman | Section 1 System overview; Section 7 ASRs and utility tree; final editing | Sections 4, 5 |
| Layal Ahmed Moussa | Section 2 Stakeholders and concerns; references | Sections 1, 3 |
| Mark Hany Elia | Section 3 System context diagram and diagram sources; Section 6 Constraints | Sections 2, 7 |
| Sara Raef Abdelsalam | Section 4 Functional requirements; QAS-1, QAS-2 (usability) | Section 6 |
| Youssef Sameh Elbendary | Section 5: QAS-3 to QAS-8 (security, availability, performance, modifiability, interoperability) | Section 4 |

## References


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